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Editor-in-Chief
IET Generation, Transmission & Distribution

Dear Editor,

We submit for your consideration the manuscript:

Adaptive Forgetting Factor for Drifting Inverter-Based Resource Penetration in Model-Based Protection Systems

for publication in *IET Generation, Transmission & Distribution*.

Context. The Self-Adaptive Model-Based Protection System (SAMBPS) uses a Levenberg–Marquardt zone-model estimator to adapt relay settings to the instantaneous IBR current ratio k_{ibr} . Existing SAMBPS publications assume k_{ibr} is approximately stationary within the estimation window. In practice, solar PV follows a diurnal ramp, wind generation executes rapid fluctuations, and topology switching creates step discontinuities — causing k_{ibr} to drift on timescales of seconds to hours. A fixed forgetting factor ($\lambda = 0.995$) creates a 63 ms tracking lag that degrades protection detection probability by up to 16% during drift.

Key contributions.

1. **CUSUM drift detector.** A one-sided CUSUM applied to successive LM estimates $\hat{k}_{ibr}$ detects drift within 3 cycles (median) for all four drift archetypes, with false-alarm rate $< 2 \times 10^{-4}$ per cycle under noise-only conditions.
2. **Logistic λ -adaptation law.** A closed-form law interpolates between $\lambda_{\min} = 0.970$ (effective window 33 samples, fast tracking during drift) and $\lambda_{\max} = 0.995$ (200 samples, low noise during stationary conditions), driven by the estimated drift rate $\dot{r}_k$.
3. **Stability proposition.** A formal proof that $\kappa_n(\mathbf{J}) < 30$ is preserved for all $\lambda_k \geq \lambda_{\min} = 0.970$, guaranteeing LM convergence and the SAMBPS Stage-2 veto integrity under all credible drift conditions.
4. **Monte Carlo validation.** 20 000 trials across four drift scenarios (solar diurnal, cloud transient, wind random walk, topology step) confirm: tracking RMSE reduced $3.4\times$ (0.062 to 0.018 pu); P_{FA} reduced to ≤ 0.002 ; $P_D \geq 0.981$ vs. 0.789–0.862 for fixed λ .

Relationship to companion papers. This is gap paper G2 in the SAMBPS series. The SAMBPS framework is developed in companion papers submitted to *IEEE Transactions on Smart Grid* (unified architecture, J4) and *IEEE Transactions on Power Delivery* (87L algorithm, J1; SEL-411L hardware validation, G1). This paper addresses a distinct open problem — the non-stationary k_{ibr} environment — that arises between fault events and is not covered in the fault-scenario validation of the companion papers.

Suggested reviewers.

- Prof. Ali Hooshyar, University of Toronto (IBR protection, adaptive relay settings)
- Prof. Wilsun Xu, University of Alberta (power quality, IBR system identification)
- Prof. Geza Joos, McGill University (power electronics, inverter-grid interaction)
- Prof. Lennart Ljung, Linköping University (system identification, recursive estimation)

The manuscript has not been submitted elsewhere and all authors have approved the final version.

Sincerely,

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